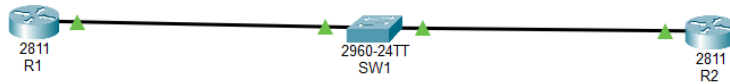


Practical Week 5

Topology



Task 28

Display Name	R1
Hostname	R1

Task 29

Display Name	R2
Hostname	R2

Task 30

Display Name	SW1
Hostname	SW1

Task 31

IP Configuration	
IPv4 Address	10.10.10.1
Subnet Mask	255.0.0.0

Task 32

IP Configuration	
IPv4 Address	10.10.10.2
Subnet Mask	255.0.0.0

Task 33

```
SW1(config)# interface vlan 10
SW1(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10

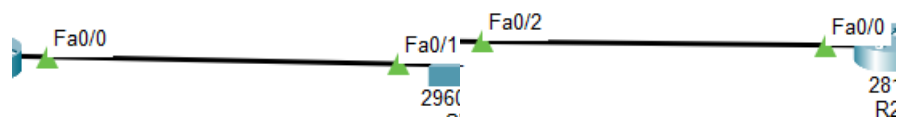
SW1(config-if)# ip address 10.10.10.10 255.0.0.0
SW1(config-if)# ip default-gateway 10.10.10.2
```

Task 34

```
SW1# ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

Task 35



Task 36

```
R1# show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID         Local Intrfce Holdtme  Capability  Platform  Port ID
SW1                Fas 0/0      136         S           2960      Fas 0/1
```

Task 37

```
R1(config)# no cdp run
R1(config)#interface fa0/1
R1(config-if)#no cdp enable
```

Task 38

```
Enter configuration for
R1(config)# no cdp run
R1(config)# cdp run
```

Task 39

```
R1#show cdp neighbors
% CDP is not enabled
```

Task 40

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    10.10.10.2     YES manual up              up
FastEthernet0/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
```

Task 41

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    10.10.10.2     YES manual administratively down down
FastEthernet0/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
```

Task 42

```

R2# configure terminal
Enter configuration commands, one per line.
R2(config)# interface fa0/0
R2(config-if)# duplex full
R2(config-if)# speed 100

R2# show interface fa0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 00e0.a3b2.1001 (bia 00e0.a3b2.1001)
  Internet address is 10.10.10.2/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    4 packets input, 512 bytes, 0 no buffer
      Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    4 packets output, 512 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out

```

Task 43

```

SW1(config)#interface fa0/2
SW1(config-if)# duplex half

SW1# show interface fa0/2
FastEthernet0/2 is down, line protocol is down (disabled)
  Hardware is Lance, address is 0090.0ca3.a902 (bia 0090.0ca3.a902)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
      Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
    0 output errors, 0 collisions, 10 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out

```

Task 44

```

SW1(config)# interface fa0/2
SW1(config-if)# duplex full

```

Task 45

```
SW1(config)#interface fa0/2
SW1(config-if)# duplex full
SW1(config-if)# speed 10 M
```

```
SW1#show interface fa0/2
FastEthernet0/2 is down, line protocol is down (disabled)
  Hardware is Lance, address is 0090.0ca3.a902 (bia 0090.0ca3.a902)
  BW 100000 Kbit, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 10Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
    0 output errors, 0 collisions, 10 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
```

Task 46

```
R2# show interface fa0/0
FastEthernet0/0 is up, line protocol is down (disabled)
  Hardware is Lance, address is 00e0.a3b2.1001 (bia 00e0.a3b2.1001)
  Internet address is 10.10.10.2/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    4 packets input, 512 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    4 packets output, 512 bytes, 0 underruns
    0 output errors, 0 collisions, 2 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
```

Task 47

When a duplex mismatch is set between the two, the interface of R2 would stop working because its speed is slow or its duplex is half or empty.